Design, Development, and Analysis of a Homemade Stellar Spectrograph

(A&A II Term Paper Presentation)
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A **spectrograph** separates light into different wavelengths and records how the intensity varies across the spectrum.

A **stellar spectrograph** is a specialized spectrograph optimized for observing stars and other stellar objects.

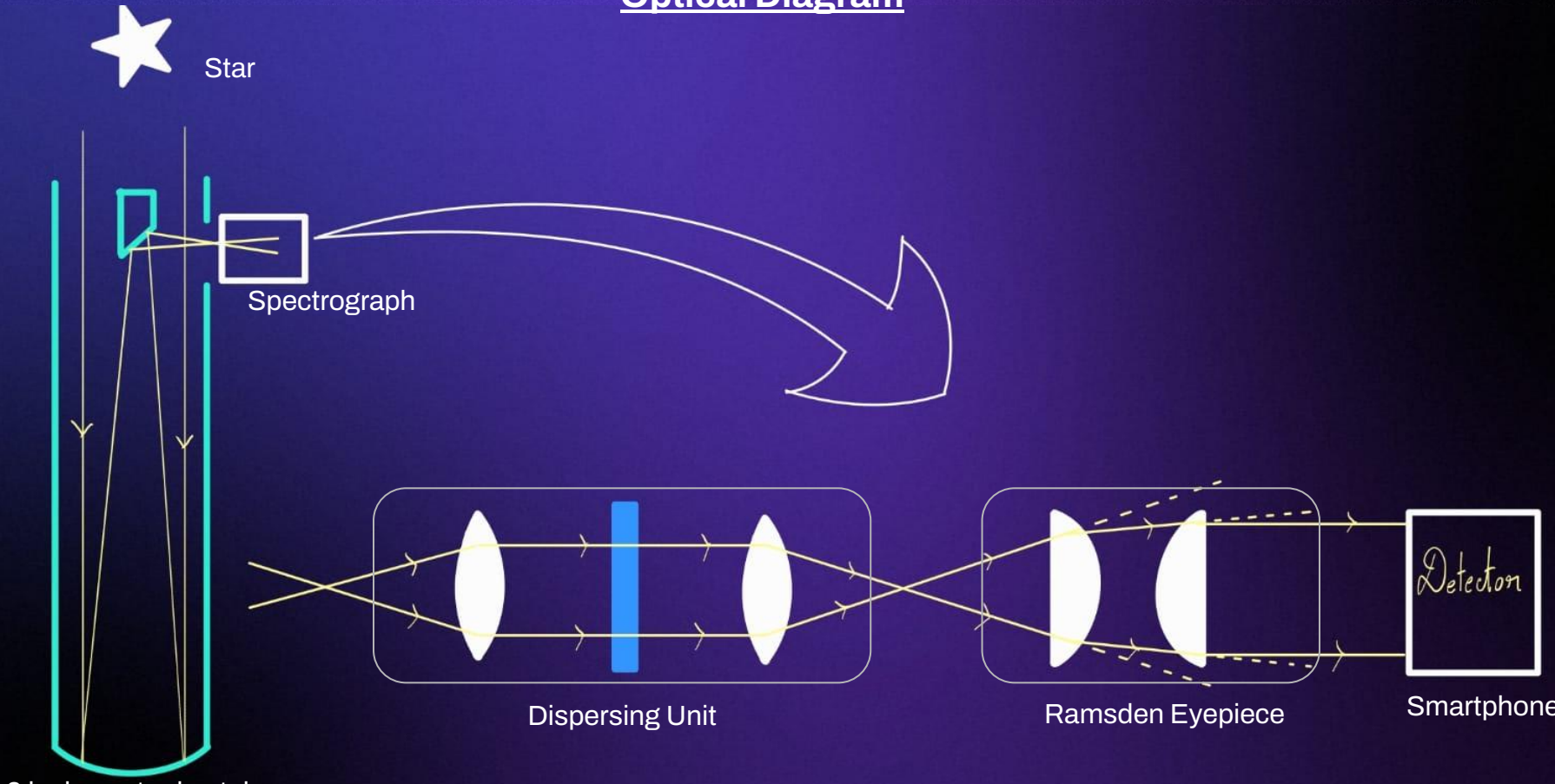
In this project, I designed, constructed, and analyzed a simple stellar spectrograph.

1. Design & Construction
2. Observation
3. Data analysis and calibration
4. Limitations

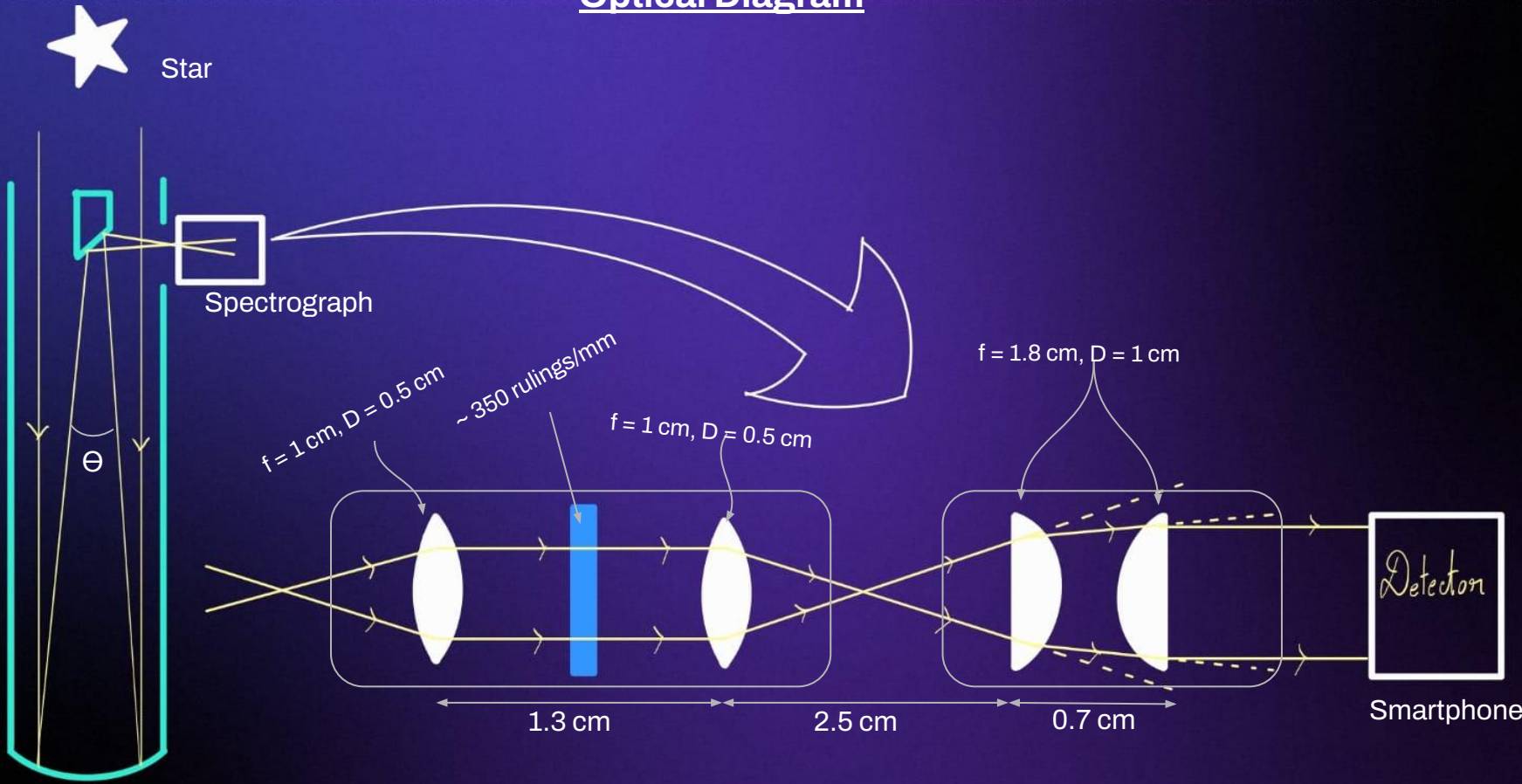
1. Design & Construction

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Optical Diagram



Optical Diagram





The Telescope I used

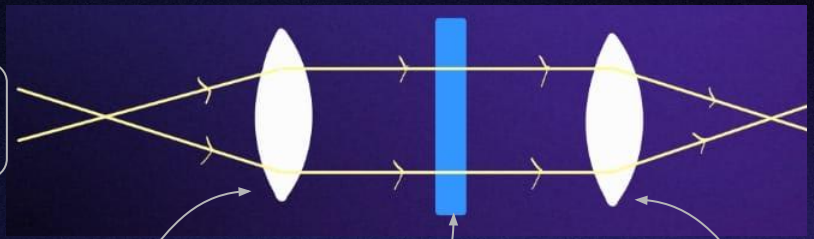


The spectrograph
(compared to my hand)

Some Numbers:

Throughput:

Dispersing Unit



Telescope beam Θ: 0.169 rad

Field lens Θ: 0.5 rad

Exit lens Θ: 0.5 rad

Each surface has T = 90%
So in total for a lens, T = 81%

Total transmission (including all the orders) is ~70%

Each surface has T = 90%
So in total for a lens, T = 81%

- Telescope Θ is smaller than field lens Θ. So all the light be allowed to enter.
- The dispersing unit has an optical throughput of approximately 46%.

- Therefore, the final throughput at the output of the dispersing unit is approximately 46% of the collected telescope light.

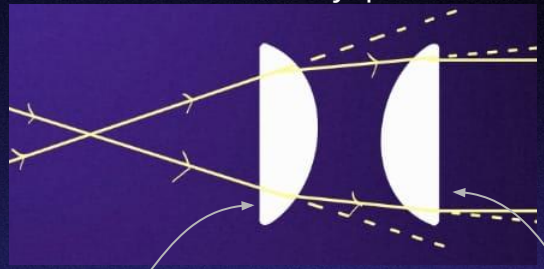
Some Numbers:

Throughput:

Dispersing unit incoming beam Θ: 0.169 rad

Eyepiece Θ: 0.67 rad (Position dependent)

Ramsden Eyepiece



Each surface has T = 90%
So in total for a lens, T = 81%

Each surface has T = 90%
So in total for a lens, T = 81%

- Eyepiece Θ is greater than dispersing unit exit beam Θ. So the eyepiece accepts the full output beam of the dispersing unit.
- The eyepiece unit has an optical throughput of approximately 65.6%.

Therefore, the final throughput at the output of the eyepiece unit is approximately
(0.46 * 0.656) ≈ 0.302 ≈ 30.2% of the collected telescope light.

Some Numbers:

Throughput:

- The optical throughput at the eyepiece output is approximately 30%.
- The grating disperses light into multiple diffraction orders, and only one principal first-order spectrum is used.
-
- The total fraction of light in that order can be calculated using the formula below. Taking the duty factor “f” = 0.5, the fraction of light in one principal maxima of order (1, 0) is 2.53%

$$\eta_{mn} = f^4 \,\mathrm{sinc}^2(m\pi f)\,\mathrm{sinc}^2(n\pi f)$$

Therefore, the final spectral throughput is approximately 0.76% of the collected telescope light.

Which is about 1 in 130 photons collected by the telescope reaches to the camera in the principle first order.

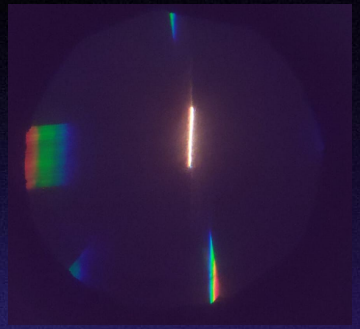
Some Numbers:

Field of View : For simplicity, the long exposure images are used to calculate the FoV.

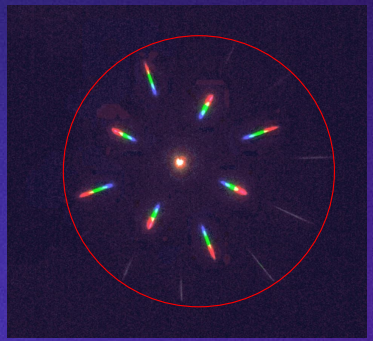
The field of view is about 4’ in diameter

Magnification: The human eye can see about 50 degrees. So using that,

The effective apparent magnification is approximately 750x



Modified FoV
(exp: 8sec)



Original version FoV

Some Numbers:

Field of View : For simplicity, the long exposure images are used to calculate the FoV.

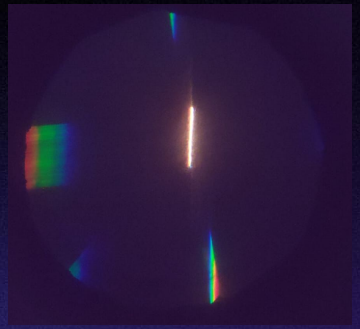
The modified version has FoV about 2’

The original one has FoV about 4’

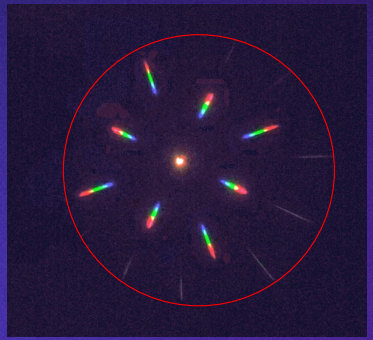
Magnification: The human eye can see about 50 degrees. So using that,

The effective apparent magnification of the modified one is approximately 1500x

The magnification in the original one is about 750x



Modified one FoV (exp: 8sec)



Original version FoV

1. Design & Construction

2. Observation

3. Data analysis and calibration

4. Limitations

For the observations, I traveled to Ghatghar, Maharashtra, and collected the data from a nearby hilltop.



TIFR NIGHT SKY
vs
GHATGHAR NIGHT SKY



For the observations, I traveled to Ghatghar, Maharashtra, and collected the data from a nearby hilltop.

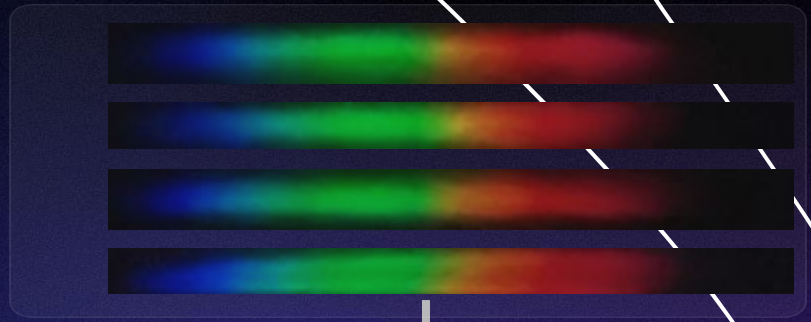


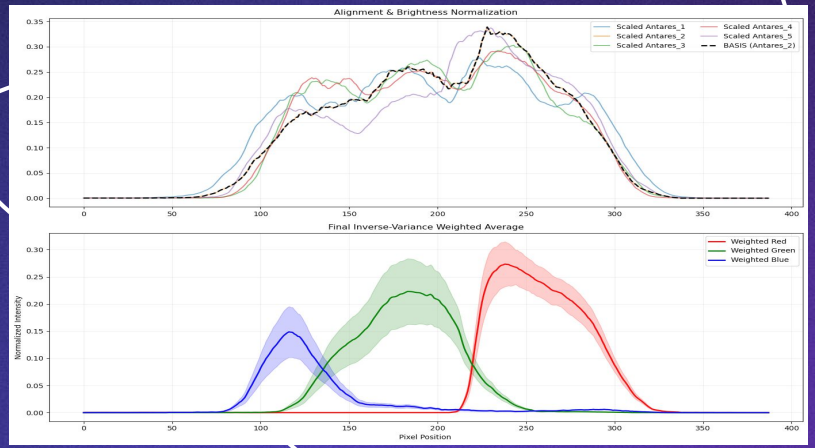
1. Design & Construction
2. Observation
3. **Data analysis and calibration**
4. Limitations

Data Analysis:

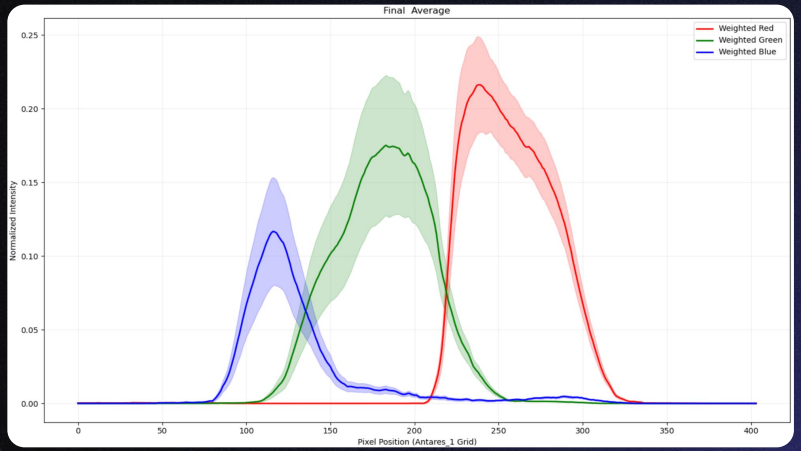


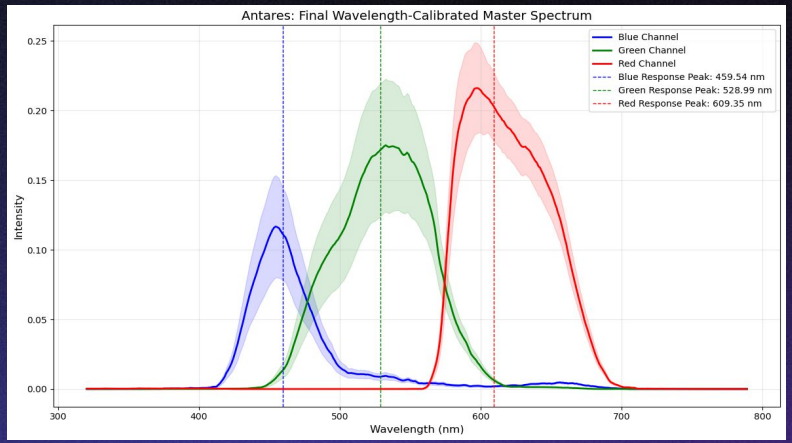
Raw spectrum of Antares,
a Red giant which is ~700 times
larger than the sun

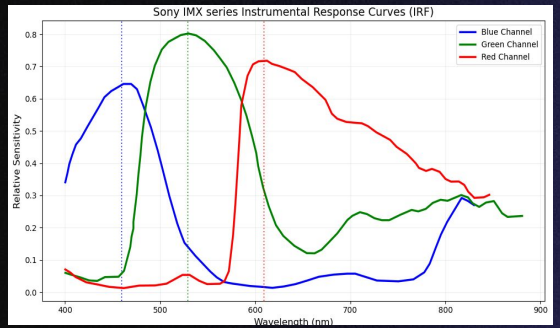




Data Analysis:







Used the **peak position** of the spectra and the Instrumental Response Function to calibrate the spectrum.

Spectral Resolution:

At least 25 nm

(Based on grating resolution and FWHM of the star in the image)

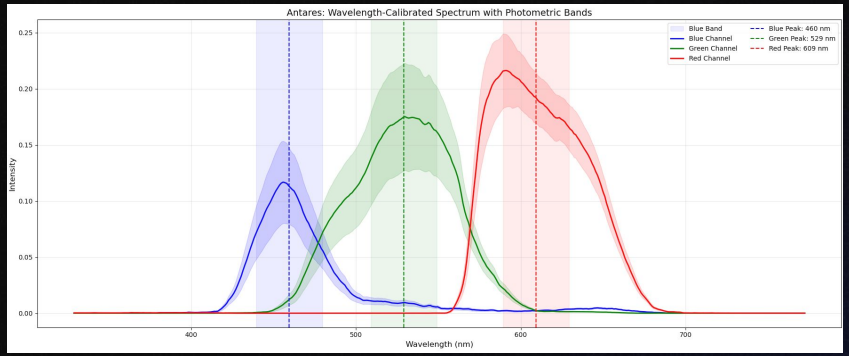
Data Analysis:

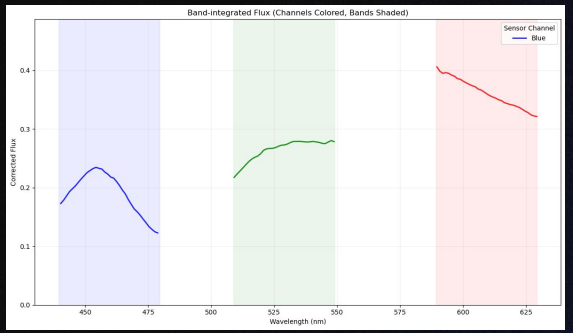
So, from the resolution it is clear that it will not work great as a spectrograph.

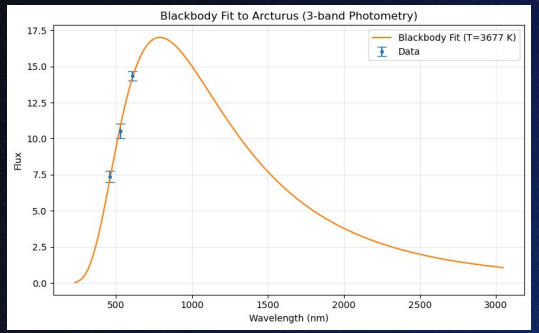
But wait !!
It can be used as a photometer instead.

Data Analysis:

Antares:







Photometric Bands

- **Band 1: 439.5 - 479.5 nm**
- **Band 2: 509.0 - 549.0 nm**
- **Band 3: 589.4 - 629.4 nm**

Temperature Analysis

Literature Value:

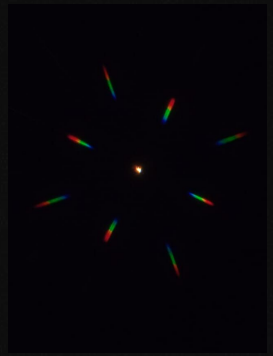
~3500 K

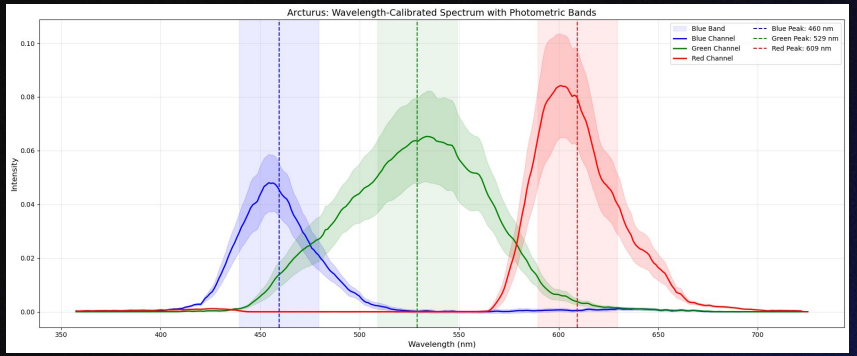
Obtained Value:

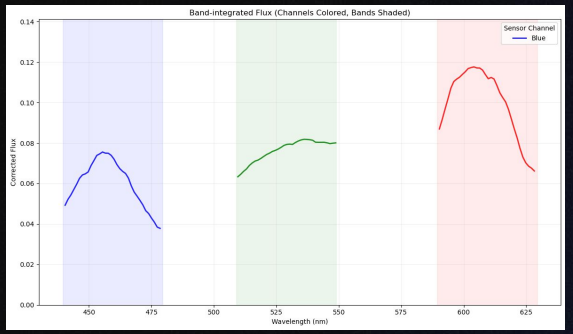
3677 ± 101 K

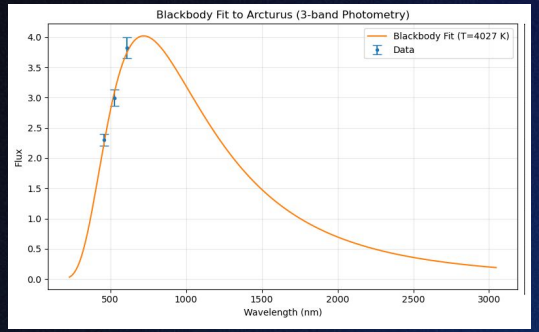
Data Analysis:

Arcturus:









Photometric Bands

- **Band 1: 439.5 - 479.5 nm**
- **Band 2: 509.0 - 549.0 nm**
- **Band 3: 589.4 - 629.4 nm**

Temperature Analysis

Literature Value:

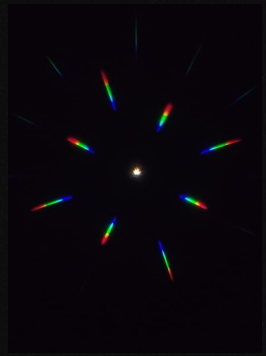
~4286 K

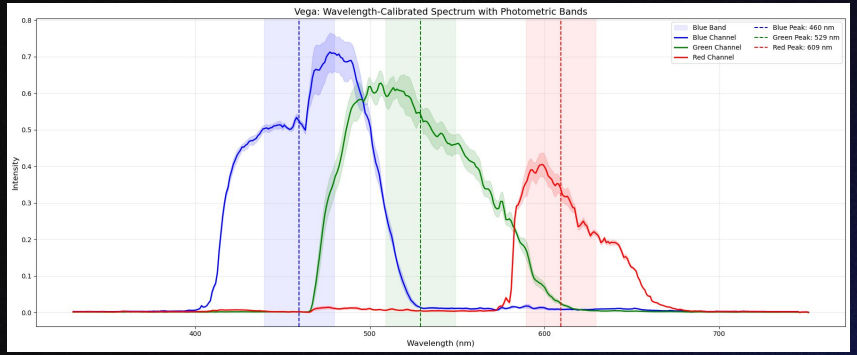
Obtained Value:

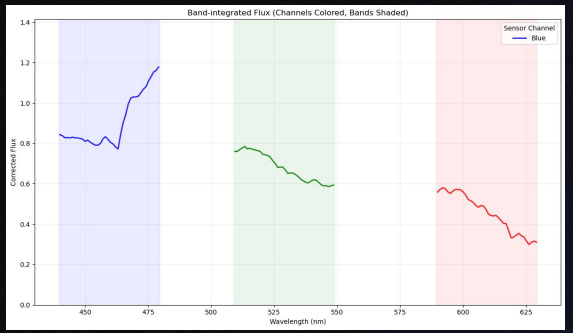
4027 ± 130 K

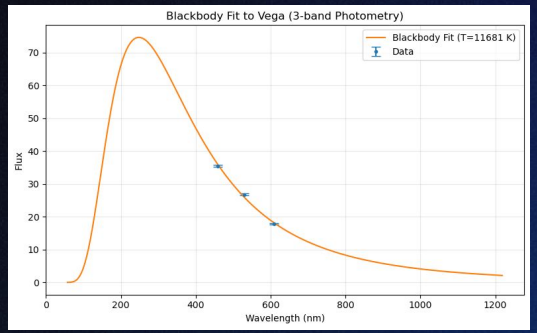
Data Analysis:

Vega:









Photometric Bands

- **Band 1: 439.5 - 479.5 nm**
- **Band 2: 509.0 - 549.0 nm**
- **Band 3: 589.4 - 629.4 nm**

Temperature Analysis

Literature Value:

~9800 K

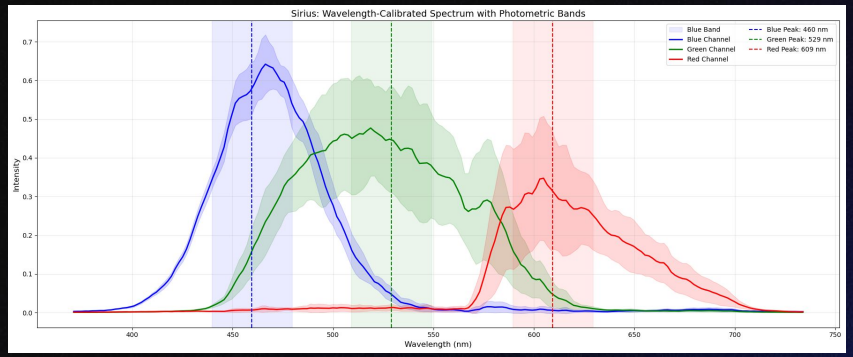
Obtained Value:

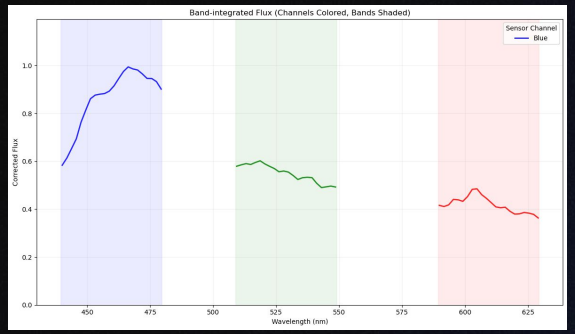
11681 ± 323 K

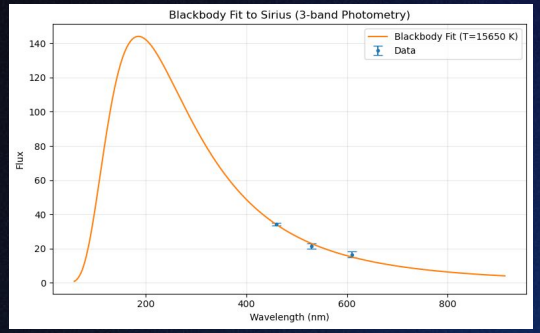
Data Analysis:

Sirius:









Photometric Bands

- **Band 1: 439.5 - 479.5 nm**
- **Band 2: 509.0 - 549.0 nm**
- **Band 3: 589.4 - 629.4 nm**

Temperature Analysis

Literature Value:

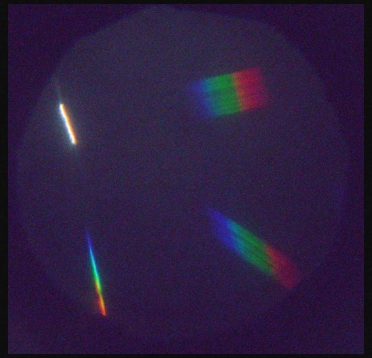
~9940 K

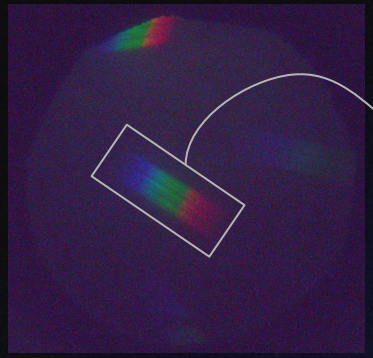
Obtained Value:

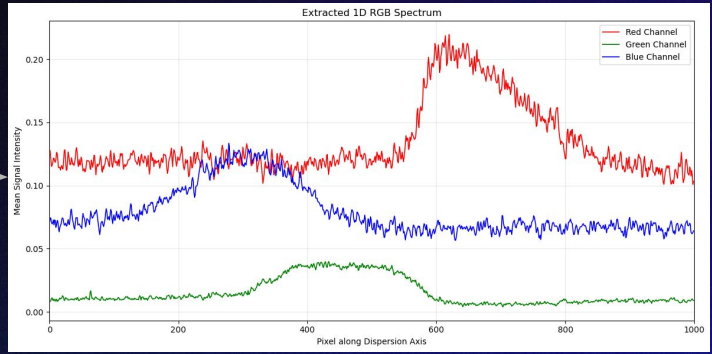
15650 ± 3735 K

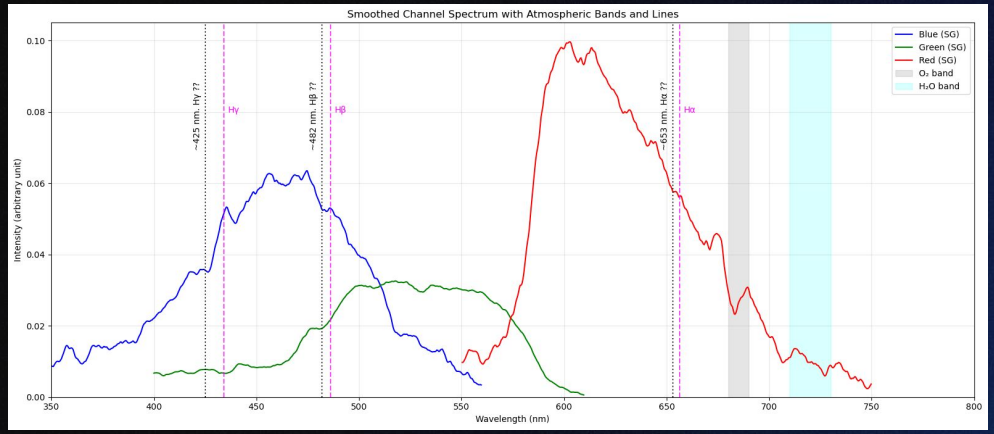
Data Analysis:

Vega (High res):









Modified the dispersing unit

Also utilized earth's rotation to get more data.

That gave $\Delta\lambda$ ~ 10 nm

Might have detected some line features.

1. Design & Construction
2. Observation
3. Data analysis and calibration
4. **Limitations**

Limitations:

1. Resolution is poor ($\Delta\lambda$ ~ 10 to 25 nm)
2. Smartphone takes JPEG images. The internal image-processing pipeline is not publicly available.
3. The exact response curve of the smartphone sensor is not available as it is not open source by the phone company. So the flux calibration is not very accurate.
4. Smartphone takes 3 channel images instead of one. So stitching the spectras are headache.
5. I haven't used any precision measurements to build this. So the numbers are just approximate values.
6. To calibrate wavelengths, I used the response curve of a sensor of same IMX series. So the wavelength calibration Is also not that accurate.
7. Mechanical instability introduced alignment drift and image distortion during long exposures.
8. The lenses have scratches. That will also affect the light output.

Nevertheless, the system performs reasonably well for a DIY instrument.

So I am happy :)

Thank you!